

Decibel Calculations

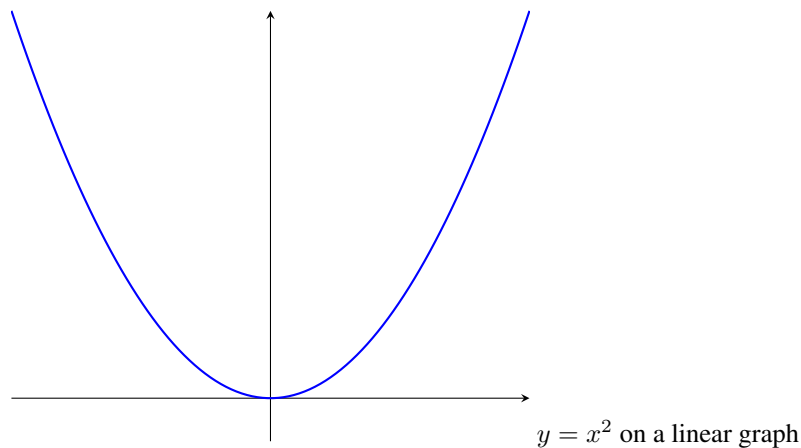
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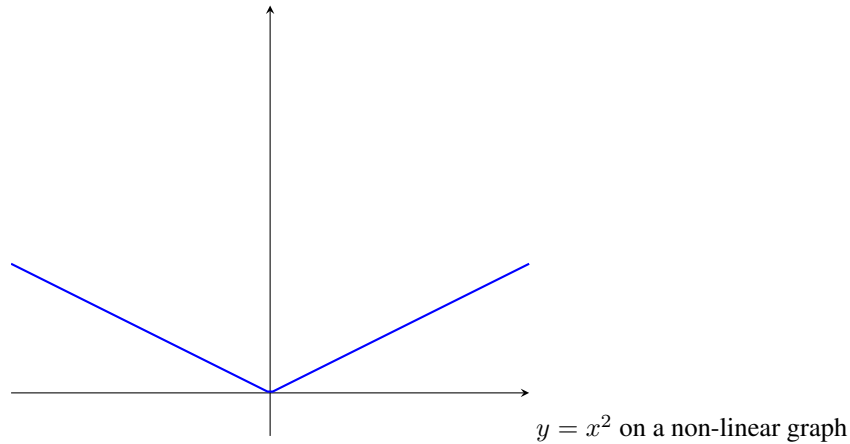
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1 Introduction

Decibel calculations are fun. In short what we are doing is working between two numerical systems. In simple terms, a linear system and a non-linear system. More specifically, a euclidean system and a logarithmic system. I trust you are familiar with linear coordinate systems and will only say that an example is your standard cartesian x - y coordinate graph.

Lets talk about logarithmic or more generally first, non-linear coordinate systems. A non-linear coordinate system is a coordinate system in which the lines or axis are weighted or in mathematical words curved. To look at this lets look at the graph of $y = x^2$ in two coordinate systems.





These two graphs are the same picture in different coordinate systems. In the first graph both the x and y axis count linearly, 1, 2, 3, ... but in the second graph the y axis counts by squares. That is to say 0, 1, 4, 9, 16, ... so the resulting image appears linear to the observer as the geometry of the graph itself has changed.

Further notice the exact change in our example is we have chosen a non-linear system which inverts the function being graphed. This can be seen even more clearly but realizing that if the x axis is counting linearly and the y axis is counting by squares that's the same as the x axis counting by square roots and the y axis counting linearly.

The point I am trying to drive home is we can perform operations on our coordinate systems themselves which results in changes in geometry. More these operations behave as then would acting within the coordinate system, thus inverses are still inverses.

So now we can see another relationship. When the inverse of the graphed function is the x axis in the coordinate system the result is a linear picture.

Now I hear you! Your saying well why do I care! This seem like unimportant weird math stuff. But this is what decibel calculations are doing! With decibel calculations we are looking at the exponential relation of orders of magnitude as a linear picture. And with that lets dive into it!

2 Definitions and Theorems

We are going to recall logarithmic and exponential math here rather than do a typical definitions section and the practice with examples.

Exponential

Let $b \in \mathbb{R}$. An exponential function $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function of the form

$$f(x) = b^x$$

There is a pattern that follows from the definition regarding exponential functions. This pattern is that if we take any two consecutive terms, the preceding term is a factor of the base b of the following term. To see this look at the quick example $f(x) = 2^x$ which has terms:

$$2^0 \quad 2^1 \quad 2^2 \quad 2^3 \quad 2^4$$

...

$$1 \quad 2 \quad 4 \quad 8 \quad 16$$

...

as you can see in this case the base is 2, so each time we see the next term double. If the base is 3 it triples, if its $\frac{1}{4}$ it quarters.

The reason for bringing this up is this, logarithmic functions are defined to be the inverse functions of exponential functions. This means their behavior is to undo the pattern I have just described. But why do we care? Well think how orders of magnitude grow. Each order of magnitude is a multiple of 10! Thus it is truly exponential! And when we take log base 10, we effectively linearize this relation.

Lets now formally define logarithmic functions and play with some examples of how to work between linear and decibels. And that word linear in a weird spot.

Logarithmic functions

Let $b \in \mathbb{R}$ and $f(x) = b^x$ be an exponential function. Then we define the logarithmic function as f^{-1} and write

$$f^{-1}(x) = \log_b(x)$$

Thus we have the property

$$x = \log_b(a) \iff b^x = a$$

Now to define decibels specifically.

Decibels

dB is decibels and P is something we want to measure a ratio in decibels, maybe power, maybe something else.

$$\text{dB} = 10 \log_{10}\left(\frac{P_{out}}{P_{in}}\right)$$

3 Examples

Looking at the example we want power out of 750 watts and we want a gain of 5 dB how would we go about finding our required power in?

First lets set up a logarithmic expression.

$$5 = 10 \log_{10} \left(\frac{750}{P_{in}} \right)$$

Now lets use our understanding of exponential and logarithmic functions to solve for P_{in} .

$$\begin{aligned} \frac{5}{10} &= \log_{10} \left(\frac{750}{P_{in}} \right) \\ 10^{\frac{1}{2}} &= 10^{\log_{10} \left(\frac{750}{P_{in}} \right)} \\ &= \frac{750}{P_{in}} \end{aligned}$$

so

$$P_{in} = \frac{750}{\sqrt{10}} \text{ watts}$$

Looking at one more example say we have an antenna with power in of 250 watts, and power out of 800 watts. then lets calculate the decibel gain.

$$\begin{aligned} \text{dB} &= 10 \log_{10} \left(\frac{P_{out}}{P_{in}} \right) \\ \text{dB} &= 10 \log_{10} \left(\frac{800}{250} \right) \\ &\approx 5.05 \text{ dB Gain} \end{aligned}$$

For more look at the identities sheet to see more properties of exponential and logarithmic functions to better use these skills.